KHUSNIL ARIFANDI PURNAMA

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Data-driven Physics graduate from ITB with a strong foundation in quantitative modeling and analytics. Proven track record in translating complex datasets into strategic business recommendations, combining rigorous analytical problem-solving with hands-on operational leadership to co-create actionable solutions for clients and stakeholders.

EDUCATION

Institut Teknologi Bandung (ITB) Bandung, Indonesia

- *Bachelor of Science in Physics* July 2022 – October 2026
- Undergraduate Thesis**:** "*Capillary Drainage Simulation Based on Pore Morphology Method for Determination of Irreducible Water Saturation & Hydraulic Flow Unit.* Developed a complex computational and quantitative model to simulate physical interactions within digital rock samples, demonstrating advanced capability in processing multidimensional data to solve intricate structural problems.
- Relevant Coursework: Simulation & Physical Modelling, Data Visualisation & Image Processing, Statistical Analysis, Project Management, Econophysics & Sociophysics

PROFESSIONAL EXPERIENCE & PROJECT

PT Djarum Kudus, Indonesia

Quality Control Internship Jul 2025 – Sep 2025

- Applied analytical & statistical thinking to evaluate production stability using X-S Control Chart via Python & Excel, identifying systematic shift variations to standardize collaborative work methods.
- Excecuted daily quality inspections on cigarettes & calibrated the QTM measurement machine to ensure data accuracy, while evaluating sampling methodologies (n=50 vs n=10) to design a proactive defect-prevention dashboard & presenting data-driven insights to the Plant Manager & Head of Production.

Coding Camp 2025 Dicoding Powered by DBS Foundation

Machine Learning Cohort Feb 2025 – Jun 2025

- Architected end-to-end data science pipelines and deployed production-ready machine learning models through a rigorous 900+ hour ML Engineering program.
- *Car Recommendation System:* Engineered a machine-learning recommendation system for vehicle selection, delivering clean, modular, and well-documented code to drive scalable collaborative development
- *Capstone Project-SmartRecruiter:* Collaborated cross-functionally to build 'SmartRecruiter', an NLP-driven full-stack web application integrating ML models to automate resume screening and candidate scoring process.

Others:

- *Stock Exchange Recommendation Website*: This platform is an institutional equity research system for the Indonesia Stock Exchange that integrates multi-engine Machine Learning consensus predictions, official IDX foreign capital flow tracking, comprehensive technical-fundamental analysis, and calibrated risk management execution. https://stock-market-recommendation-silk.vercel.app/

LEADERSHIP EXPERIENCE

HIMAFI ITB (Physics Student Association) Bandung, Indonesia

Staff – Alumni Relation & Career Development June 2024 – April 2025

- Spearheaded the development of a comprehensive alumni database and curated targeted job opportunities, effectively connecting physics graduates with strategic roles in top professional services firms that actively recruit hundreds of university graduates annually.

Apres ITB! Bandung, Indonesia

Head – Operation & Asset Management Aug 2024 – Jul 2025

- Spearheaded the organization's first-ever musical equipment rent ecosystem, transforming idle assets into a sustainable business model that successfully generated 30% of the total annual revenue.
- Led an 8-member cross-functional team to execute MUSTANG 2025, a large-scale campus music festival, ensuring 100% equipment availability and optimizing the end-to-end logistical supply chain.

SKILLS, LANGUAGES & INTEREST

- **Data Science & Analytics:** Machine Learning, Time-Series Forecasting, Quantitative Modelling, Statistical Analysis, Data Visualisation
- **Programming & Tools:** Python (Scikit-Learn, TensorFlow), Excel (Advanced Data Management & Reporting)
- **Business & Finance Acumen:** Financial Statement Analysis, P&L & Cash-Flow Analysis, Business & Market Analysis, Market Research.
- **Languages:** Indonesian (Native), English (Professional/ Toefl ITP Score 507)